YAMLess Analysis Report

pos-elera-sales-compliance

Date: 2026-11-10

Repo:

App Summary

Field	Value	Confidence

Detected Dependencies

Dependency	Detected	Notes

Dependency Detection Gaps

The static analyzer correctly detects Redis (cache flag). However, two additional runtime dependencies are present in the secret surface and were **not detected** by the static scanner:

• **MongoDB** — `MONGODB_URI` and `SPRING_DATA_MONGODB_URI` appear in the secrets list. MongoDB is a NoSQL document store; the current dependency scanner only classifies `postgres` and `mysql` SQL databases. As a result, `storage: false` and `database: null` are returned despite an active MongoDB connection being present.

• **RabbitMQ / AMQP** — `SPRING_RABBITMQ_PASSWORD` and `AMQP_BROKER_PASSWORD` indicate a live RabbitMQ or AMQP message broker connection. The scanner's `queue` flag maps to Azure Service Bus; it does not pattern-match RabbitMQ/AMQP package or env-var signals, so `queue: false` is returned.

Both gaps would be caught by the **LLM gap-filler** when run with `ANTHROPIC_API_KEY` set (see Recommendations).

Run 1 — Azure Container Apps

Command: `yamless generate pos-elera-sales-compliance --target container-apps --no-llm --validate`

Generated Infrastructure

35 Terraform files written to `outputs/pos-elera-sales-compliance-infra/`

Cost Estimate

Resource	Monthly (low)	Monthly (high)	Notes

The Container App cost range reflects consumption-based scaling (1–2 replicas). Redis P1 ($370/mo) dominates the cost profile. The P1 SKU is selected by the golden profile for production compliance workloads where high availability and in-memory performance are required.

Checkov Results

30 / 31 checks passed

Result	Count

CKV_AZURE_43 — Storage account name does not follow naming rules

• **Severity:** MEDIUM

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Run 2 — Azure AKS

Command: `yamless generate pos-elera-sales-compliance --target aks --no-llm --validate`

Generated Infrastructure

35 Terraform files + **8 Kubernetes manifests (.yaml)** written to `outputs/pos-elera-sales-compliance-aks-infra/`

Kubernetes manifests include: `namespace.yaml`, `service-account.yaml`, `secret-provider-class.yaml`,

`deployment.yaml`, `service.yaml`, `ingress.yaml`, `hpa.yaml`, `pdb.yaml`

Cost Estimate

| Resource | Monthly | Annual | Notes |
|----------

AKS cost is fixed regardless of replica count, since the node pool runs continuously. Appropriate for workloads with steady, predictable traffic.

Checkov Results

48 / 49 checks passed

Result	Count

CKV_AZURE_43 — Storage account name does not follow naming rules

• **Severity:** MEDIUM

• **Asses
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Run Comparison

	Container Apps	AKS

Target Recommendation

AKS ($985/mo fixed) is preferred for `pos-elera-sales-compliance`. POS compliance services typically run at steady load with predictable traffic patterns — AKS fixed pricing is more cost-predictable than the Container Apps variable range ($648–$1,042/mo). AKS also provides a full Kubernetes control plane, enabling fine-grained network policies, pod security standards, and workload identity integration appropriate for a compliance workload. The 8 K8s manifests generated by the AKS run enforce topology spread constraints, pod disruption budgets, and HPA stabilization windows out of the box.

Secrets Breakdown

Total secrets detected: 43

Ping Identity / OIDC (17 secrets)

Secret	Purpose

This service integrates with both **PingFederate** (enterprise on-premises SSO) and **PingOne** (cloud SSO) simultaneously, suggesting a dual-IdP OIDC setup during a migration or for different client types.

MongoDB (2 secrets)

Secret	Purpose

RabbitMQ / AMQP (3 secrets)

Secret	Purpose

Kount Fraud Detection (2 secrets)

Secret	Purpose

ELERA POS Platform (3 secrets)

Secret	Purpose

Redis Cache (2 secrets)

Secret	Purpose

Other External Services (remaining secrets)

Secret	Purpose

Analyst Notes and Recommendations

1. Re-run with `--llm` to Add MongoDB + RabbitMQ Modules

The static analyzer returned `confidence: high` for runtime, secrets, and shape. However, two significant runtime dependencies — MongoDB and RabbitMQ — were **not detected** because the static scanner does not pattern-match NoSQL databases or AMQP brokers. Running the full pipeline with `ANTHROPIC_API_KEY` set will invoke the LLM gap-filler, which reads source files and infers these dependencies from context.

```bash
yamless
```

Adding MongoDB and RabbitMQ would materially change the generated infrastructure (additional Azure Cosmos DB and Service Bus modules), and would increase estimated cost.

2. Add MongoDB Module to Generated Terraform

Once the gap-filler detects MongoDB, the renderer should emit an **Azure Cosmos DB for MongoDB** module.

Cosmos DB provides MongoDB wire-protocol compatibility with Azure-native management, private endpoints, and automatic indexing.

Expected additional cost: ~$25–$100/mo (serverless tier) or ~$600+/mo (provisioned throughput).

3. Add RabbitMQ / AMQP Module

RabbitMQ is not a native Azure service. Options for the generated Terraform:

• **Azure Service Bus** — if the team can migrate AMQP clients to AMQP 1.0 (Service Bus is AMQP 1.0 compatible; RabbitMQ is AMQP 0-9-1). This is the preferred cloud-native path and is already supported by the yamless `servicebus` module.

• **Self-h

Consider opening a backlog item to add RabbitMQ/AMQP detection in `analyzer/dependencies.py` (pattern-match `spring-rabbit`, `amqp`, `amqplib` package names and `RABBITMQ`/`AMQP` env-var prefixes).

4. PingFederate + PingOne Dual-IdP Pattern

The 17 Ping-related secrets indicate simultaneous integration with both PingFederate (enterprise on-premises SSO) and PingOne (cloud SSO). This is a recognized migration pattern. All Ping secrets are correctly identified and will be stored in Azure Key Vault. No action required on the infrastructure side, but the security team should review whether duplicate issuer URIs (`PING_AUTH_ISSUER_URI` vs `PING_FED_AUTH_ISSUER_URI`) represent intentional dual-validation or a configuration redundancy.

5. Kount Fraud API

`KOUNT_API_KEY` and `KOUNT_BASE_URL` are correctly detected as secrets and will be stored in Key Vault. Kount is a third-party fraud detection platform used in POS and payment compliance flows. No additional infrastructure is needed.

6. High Secret Count (43)

43 secrets is on the high end for a single service and reflects the complexity of this microservice's external integrations: dual IdP (PingFed + PingOne), MongoDB, RabbitMQ, Redis, Kount, ELERA POS, compliance logging, and Spring Cloud Config. All 43 will be stored in Azure Key Vault under the generated workspace. Consider auditing for duplicates — `MONGODB_URI` and `SPRING_DATA_MONGODB_URI` likely point to the same host, and several `PING_*` / `PINGFED_*` variables may be redundant config aliases.

Summary

`pos-elera-sales-compliance` is a production Java 25 POS compliance microservice with a rich integration surface: dual Ping Identity SSO, MongoDB, RabbitMQ/AMQP, Redis cache, Kount fraud detection, and the ELERA POS platform. Two runs were executed against different targets.

Static analysis coverage (both runs):

• Runtime

Container Apps run: 35 .tf files, $648–$1,042/mo, 30/31 Checkov checks pass.

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The single MEDIUM Checkov failure (CKV_AZURE_43) is a false positive caused by Checkov scanning an uninstantiated storage module — present in both runs.

Recommended target: AKS for predictable fixed cost and full Kubernetes control plane.

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Generated by YAMLess — https://github.com/zafar-ahmed-kr/YAMLess